

# Willian Vieira PhD

Statistical Modeler and Data Engineer with a PhD in Ecology. I build machine learning pipelines on large-scale geospatial and remote-sensing data — from probabilistic models to the production infrastructure that runs them.

## DATA SCIENCE & ENGINEER EXPERIENCE

2025  
|  
2024  
(1 yr 9 m)

- Data Analyst**  
**Habitat**, Montreal, Canada  
Architected end-to-end R/Python ETL pipelines to deploy geospatial ML models into production environments | Built and maintained reproducible pipelines processing large-scale satellite imagery and multi-source spatial data for continental monitoring | Designed a geospatial lakehouse architecture for streaming unstructured datasets from Azure cloud storage, optimizing data retrieval for analytics and ML training | Implemented metadata-driven workflows and automated validation for training-dataset governance and pipeline reliability | Led the transition to a Unix-based production environment — Docker, CI/CD, automated setup scripts, DevOps best practices

2024  
|  
2017

- PhD Research**  
**Integrative Ecology Lab**, Sherbrooke, Canada  
Built end-to-end geospatial data pipelines to ingest, clean, and harmonize continental-scale forest-climate datasets for Bayesian hierarchical modelling | Developed reproducible HPC (High-Performance Computing) workflows (🔗), leveraging parallel processing and cluster management to execute thousands of model simulations at scale | Developed open-source software libraries (🔗, 🔗) for geospatial demographic modelling, implementing version control and automated testing | Published a **technical methods book** documenting the full computational pipeline to ensure long-term project reproducibility, along with one peer-reviewed **publication** and two preprints (📄, 📄)

2022  
|  
2020  
(part-time)

- Biostatistician**  
**Environment and Climate Change Canada** - Quebec, Canada  
Developed a cost-aware probability sampling protocol to optimize the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias correction methodologies, designing a method later adopted by other provinces | Engineered a fully automated and reproducible **geospatial pipeline** featuring automated documentation and version-controlled workflows to ensure long-term system maintainability | Produced a fully open source, reproducible **analytical workflow** with automated documentation

## EDUCATION

2024  
|  
2017

- PhD, Ecology**  
**Université de Sherbrooke** - Sherbrooke, Canada  
📖 *How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016  
|  
2015

- Masters 2, Agroecology and Resource Management**  
**Bordeaux Sciences Agro**, Bordeaux, France  
📖 *Modelling the dispersion of weed species in agricultural landscapes*

2015  
|  
2010

- BSc in Agronomy**  
**Universidade Federal de Santa Catarina**, Florianópolis, Brazil

## CONTACT

📞 (263) 381-3636  
✉ [w.vieiraw@gmail.com](mailto:w.vieiraw@gmail.com)  
in [Linkedin/WillVieira](#)  
🐙 [GitHub/WillVieira](#)  
📄 Publications

## SKILLS

**Machine Learning & Modelling:**  
Gaussian Process · Geospatial Foundation models · Bayesian inference · Hierarchical models · Uncertainty · Spatial statistics

**Programming & Database:**  
R · Python · Stan · Julia · DuckDB · PostGIS · Bash · SQL

**Geospatial & Cloud:**  
Remote sensing · ETL/ELT · Azure · AWS · Docker · Apache Arrow · STAC · COG · GeoParquet · HPC

**DevOps & Workflow:**  
Git · CI/CD · Unix · Unit testing · Make · Quarto · Jupyter

## OUTREACH

· Developed and maintained the **QCBS R Workshop Series**, reaching nearly one thousand graduate students · Taught +250 students across biology, engineering, and graduate programs (programming, statistics, ecology) · **5 published** papers & preprints with ~100 citations · Translated client needs into technical analyses and delivered clear, actionable results

## LANGUAGES

Portuguese · Native  
French · Full Professional  
English · Full Professional

CV source code hosted on 🔗  
Last updated: 2026-05-08  
📄 PDF download available